

MUSA RIOBA

Data Engineer • Author • Technical Trainer

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PROFESSIONAL SUMMARY

Data Engineer with a BSc in Actuarial Science and 6+ years of hands-on experience across freelancing, technical training, and open-source development. Author of "Production Data Science with R," an 8-chapter guide to production-grade ML pipelines, ETL orchestration, real-time fraud detection, interactive dashboards, and A/B testing frameworks. Builds systems that make data reliable, scalable, and actionable — from CDISC-standard clinical trial dashboards to national census data packages used across three countries.

SELECTED PROJECTS

TrialView Pro — Clinical Trial Monitoring Dashboard

R Shiny • Docker • CI/CD • SQLite

- Built a production-ready Phase II diabetes clinical trial monitoring system on CDISC SDTM-standard data, with role-based access control for Investigator, Monitor, and Admin roles.
- Implemented enrollment and safety monitoring (AE/SAE tables, SOC drill-down, CSV export), efficacy tracking, and longitudinal patient management.
- Added a full SQLite-backed audit trail for compliance and deployed via Docker and Shinyapps.io with a GitHub Actions CI/CD pipeline (linting, styling, data-integrity tests).

Sales Analytics Dashboard — Business Intelligence Tool

R Shiny • bslib • Plotly • Live App

- Delivered a production BI dashboard for sales analytics and executive reporting with reactive filters (region, product, time period) driving live KPI cards and Plotly charts.
- Built a sortable transactions table with styled status badges (DT) and one-click CSV export, in a fully responsive dark-theme UI.

Census Data Packages — kenyacensus, ugcensus, sacensus

R Packages • Open Source

- Published three open-source R packages providing toolkits for accessing, cleaning, and visualizing national census data for Kenya (8 census years, 47 counties, 13 datasets), Uganda (4 census years, 17 sub-regions), and South Africa (4 census years, 9 provinces).
- Built in multi-level geography support, population pyramids, growth-rate projections, and time-series comparison tooling; kenyacensus released under a CCO public-domain license.

PUBLISHED WORK

"Production Data Science with R"

Author

An 8-chapter guide to building production-grade data systems in R, covering end-to-end ML pipelines, automated ETL with targets, real-time fraud detection with Redis and isolation forests, demand forecasting (Prophet + XGBoost ensembles), customer segmentation (RFM, K-Means), marketing attribution (Markov chains), interactive Shiny dashboards, and sequential/Bayesian A/B testing frameworks — organized around reproducibility, scalability, observability, and safety.

TECHNICAL TRAINING

Technical Trainer

4 years

- Taught data analysis, R, Python, and production best practices to 100+ students and professionals.

TECHNICAL SKILLS

Languages: R (Expert), Python, SQL, Bash, HTML/CSS

Data Engineering & Pipelines: targets (ETL orchestration), ETL/ELT pipelines, data cleaning, data modeling, web scraping, API integration

Machine Learning & Analytics: tidymodels, XGBoost, Prophet, K-Means/RFM, Markov chains, isolation forests

Dashboards & Visualization: R Shiny, ggplot2, Plotly, shinydashboard, DT (DataTables)

Production & Deployment: Shinyapps.io, plumber (APIs), Redis, Git/GitHub, GitHub Actions

Databases & Cloud: PostgreSQL (RPostgres), BigQuery (bigrquery), dbplyr, AWS (learning), Docker (learning)

Experimentation & Testing: A/B testing, sequential testing, Bayesian methods, statistical modeling

Growing into: Apache Airflow, dbt, Snowflake, Spark, Kafka

EDUCATION

BSc, Actuarial Science

Karatina University